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Pass-through wiring in notched interconnect

Abstract

Embodiments of present invention provide a semiconductor structure. The semiconductor structure includes a metal level, the metal level includes a metal strip including a notch at a side of the metal strip or a pass-through inside the metal strip, wherein the notch or the pass-through is at least partially filled with a dielectric material. The metal level further includes a conductive wiring that vertically passes through the metal strip. The conductive wiring is at least partially inside the notch or inside the pass-through and is insulated from the metal strip by the dielectric material. Methods of manufacturing the semiconductor structure are also provided.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6503845	12/2002	Nallan	N/A	N/A
7354853	12/2007	Eissa	N/A	N/A
8629536	12/2013	Mina	N/A	N/A
9508642	12/2015	Bouche	N/A	N/A
9805972	12/2016	Zhang	N/A	N/A
9911651	12/2017	Briggs	N/A	N/A
9922876	12/2017	Lee	N/A	N/A
10020254	12/2017	Bao	N/A	N/A
10026687	12/2017	Lin	N/A	N/A
10515894	12/2018	Briggs	N/A	N/A
10741441	12/2019	Li	N/A	N/A
11163932	12/2020	Shao	N/A	N/A
11201112	12/2020	Cheng	N/A	N/A
2006/0180930	12/2005	Nguyen	N/A	N/A
2015/0140812	12/2014	Zope	N/A	N/A
2021/0375722	12/2020	Kim	N/A	H10D 84/038
2022/0005762	12/2021	Mignot	N/A	N/A
2022/0139863	12/2021	Song	N/A	N/A
2023/0245924	12/2022	Narkeviciute	257/760	H01L 21/32

OTHER PUBLICATIONS

Zhao et al., “Surface chemistry of thermal dry etching of cobalt thin films using hexafluoroacetylacetone (hfacH)”, ScienceDirect, Applied Surface Science, vol. 455, Oct. 15, 2018, 4 pages. cited by applicant

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Background/Summary

BACKGROUND

(1) The present application relates to manufacturing of semiconductor integrated circuits. More particularly, it relates to pass-through wiring in notched interconnect and method of manufacturing thereof.

(2) Advanced semiconductor chips contain millions or even billions of transistors that are interconnected and electrically powered to achieve certain device functionality. Current chip technology generally employs various metal lines or wirings made at the frontside of the chip, such

as various metal levels with each metal level including various metal lines or wirings in a back-end-of-line (BEOL) setting, and the metal lines or wirings are made for signal routing and/or power distribution. As new technology platform emerges, there is a developing trend of moving at least some power delivery and/or distribution network to the backside of the chip, leaving the precious real estate of frontside of the chip for signal routing.

(3) In a backside power distribution network (BSPDN), as the name implies, the wirings are mainly designed for power delivery and/or distribution, and generally there is not enough room left for making additional wirings for signal routing purpose. This poses a serious challenge when, in certain situations, some signal routing wirings need to be moved from the frontside of the chip to the backside of the chip.

SUMMARY

(4) Embodiments of present invention provide a semiconductor structure. The semiconductor structure includes a metal level, the metal level includes a metal strip, including a notch at a side of the metal strip, and a conductive wiring, and the conductive wiring is at least partially inside the notch; vertically passes through the metal strip; and is insulated from the metal strip by a dielectric material.

(5) In one embodiment, the semiconductor structure further includes a substrate and one or more transistor structures on a frontside of the substrate, where the metal level is at a backside of the substrate.

(6) In another embodiment, the metal level is a first metal level; the metal strip is a first metal strip; and the notch is a first notch, the semiconductor structure further includes a second metal level above the first metal level, the second metal level including a second metal strip including a second notch at a side of the second metal strip, where the conductive wire is at least partially inside the second notch and insulated from the second metal strip.

(7) In yet another embodiment, the metal level is a first metal level; and the metal strip is a first metal strip, the semiconductor structure further includes a second metal level above the first metal level, the second metal level including a second metal strip including a pass-through inside the second metal strip, where the conductive wire is inside the pass-through and insulated from the second metal strip.

(8) In one embodiment, the second metal strip of the second metal level is oriented to be orthogonal to the first metal strip of the first metal level, and the conductive wiring is a conductive via.

(9) In one embodiment, the notch formed at the side of the metal strip has a depth into the metal strip, wherein the depth is 50 percent or less of a width of the metal strip. In another embodiment, the metal strip has a width ranging from about 10 nm to about 100 nm.

(10) Embodiments of present invention further provide a semiconductor structure. The semiconductor structure includes a metal level, the metal level including a metal strip including a pass-through inside the metal strip, where the pass-through is at least partially filled with a dielectric material.

(11) In one embodiment, the metal level further includes a conductive wiring, the conductive wiring vertically passing through the pass-through and being insulated from the metal strip by the dielectric material.

(12) In one embodiment, the metal level is a first metal level; the metal strip is a first metal strip; and the pass-through is a first pass-through, the semiconductor structure further includes a second metal level above the first metal level, the second metal level including a second metal strip including a second pass-through inside the second metal strip, where the conductive wiring is inside the second pass-through and insulated from the second metal strip.

(13) Embodiments of present invention provide a method. The method includes providing a substrate with a frontside and a backside and at least one transistor structure on the frontside of the substrate; forming a metal level at the backside of the substrate, forming the metal level including forming a metal strip including a notch at a side of the metal strip or including a pass-through

inside the metal strip, where the notch or the pass-through is at least partially filled with a dielectric material.

(14) In one embodiment, forming the metal level further includes forming a conductive wiring that vertically passes through the metal strip, where the conductive wiring is at least partially inside the notch or inside the pass-through and being insulated from the metal strip by the dielectric material. According to one embodiment, forming the metal strip includes etching a layer of conductive material exposed by a hard-mask pattern to form the conductive wiring and the metal strip with the notch or the pass-through. According to another embodiment, forming the metal strip includes etching a layer of dielectric material exposed by a hard-mask pattern to create one or more openings in the layer of dielectric material and filling the one or more openings with a conductive material to form the conductive wiring and the metal strip with the notch or the pass-through.

(15) In another embodiment, the method further includes creating an opening in the dielectric material in the notch or in the pass-through; and filling the opening with a conductive material to form a conductive wiring, the conductive wiring vertically passing through the metal strip and being insulated from the metal strip by the dielectric material.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present invention will be understood and appreciated more fully from the following detailed description of embodiments of present invention, taken in conjunction with accompanying drawings of which:

(2) FIGS. 1A-1C to 3A-3C are demonstrative illustrations of top views and cross-sectional views of a semiconductor structure in a process of manufacturing thereof according to one embodiment of a method of present invention;

(3) FIGS. 4A-4C to 6A-6C are demonstrative illustrations of top views and cross-sectional views of a semiconductor structure in a process of manufacturing thereof according to another embodiment of a method of present invention;

(4) FIGS. 7A-7C to 10A-10C are demonstrative illustrations of top views and cross-sectional views of a semiconductor structure in a process of manufacturing thereof according to yet another embodiment of a method of present invention;

(5) FIGS. 11A-11C to 14A-14C are demonstrative illustrations of top views and cross-sectional views of a semiconductor structure in a process of manufacturing thereof according to a further embodiment of a method of present invention;

(6) FIG. 15 is a demonstrative illustration of top view of a semiconductor structure according to embodiments of present invention; and

(7) FIG. 16 is a demonstrative illustration of a flow-chart of a method of manufacturing a semiconductor structure according to embodiments of present invention.

(8) It will be appreciated that for simplicity and clarity purpose, elements shown in the drawings have not necessarily been drawn to scale. Further, and if applicable, in various functional block diagrams, two connected devices and/or elements may not necessarily be illustrated as being connected. In some other instances, grouping of certain elements in a functional block diagram may be solely for the purpose of description and may not necessarily imply that they are in a single physical entity, or they are embodied in a single physical entity.

DETAILED DESCRIPTION

(9) In the below detailed description and the accompanying drawings, it is to be understood that various layers, structures, and regions shown in the drawings are both demonstrative and schematic illustrations thereof that are not drawn to scale. In addition, for the ease of explanation, one or more layers, structures, and regions of a type commonly used to form semiconductor devices or

structures may not be explicitly shown in a given illustration or drawing. This does not imply that any layers, structures, and regions not explicitly shown are omitted from the actual semiconductor structures. Furthermore, it is to be understood that the embodiments discussed herein are not limited to the particular materials, features, and processing steps shown and described herein. In particular, with respect to semiconductor processing steps, it is to be emphasized that the descriptions provided herein are not intended to encompass all of the processing steps that may be required to form a functional semiconductor integrated circuit device. Rather, certain processing steps that are commonly used in forming semiconductor devices, such as, for example, wet cleaning and annealing steps, are purposefully not described herein for economy of description.

(10) It is to be understood that the terms “about” or “substantially” as used herein with regard to thicknesses, widths, percentages, ranges, etc., are meant to denote being close or approximate to, but not exactly. For example, the term “about” or “substantially” as used herein implies that a small margin of error may be present such as, by way of example only, 1% or less than the stated amount. Likewise, the terms “on”, “over”, or “on top of” that are used herein to describe a positional relationship between two layers or structures are intended to be broadly construed and should not be interpreted as precluding the presence of one or more intervening layers or structures.

(11) To provide spatial context to different structural orientations of the semiconductor structures shown in the drawings, XYZ Cartesian coordinates may be provided in some of the drawings. The terms “vertical” or “vertical direction” or “vertical height” as used herein denote a Z-direction of the Cartesian coordinates shown in the drawings, and the terms “horizontal” or “horizontal direction” or “lateral direction” as used herein denote an X-direction and/or a Y-direction of the Cartesian coordinates shown in the drawings.

(12) Moreover, although various reference numerals may be used across different drawings, the same or similar reference numbers are used throughout the drawings to denote the same or similar features, elements, or structures, and thus detailed explanations of the same or similar features, elements, or structures may not be repeated for each of the drawings for economy of description. Labelling for the same or similar elements in some drawings may be omitted as well in order not to overcrowd the drawings.

(13) FIGS. 1A, 1B, and 1C to FIGS. 3A, 3B, and 3C are demonstrative illustrations of top views and cross-sectional views of a semiconductor structure in a process of manufacturing thereof according to one embodiment of a method of present invention. More particularly, figures numbered with a suffix “A” are top views of the semiconductor structure, figures numbered with a suffix “B” are cross-sectional views of the semiconductor structure at a location indicated by a dashed line Y-Y, and figures numbered with a suffix “C” are cross-sectional views of the semiconductor structure at a location indicated by a dashed line X-X. It is to be noted here that similar suffixes “A”, “B”, and “C” may be applied to FIGS. 4-14, as being described below in more details.

(14) For example, embodiments of present invention provide receiving or providing a semiconductor substrate having a frontside and a backside. The frontside of the semiconductor substrate may include one or more transistor structures in a front-end-of-line (FEOL) structure. In one embodiment, a back-end-of-line (BEOL) structure with one or more metal levels may be formed at the frontside of the semiconductor substrate on top of the FEOL structure. In another embodiment, a backside power distribution network (BSPDN) structure with one or more metal levels may be formed at the backside of the semiconductor substrate. Hereinafter, embodiments of present invention may be described mainly in the context of forming the one or more metal levels at the backside of the semiconductor structure. However, a person skilled in the art will appreciate that same or similar description may be applied in forming the one or more metal levels at the frontside of the semiconductor substrate.

(15) In forming the one or more metal levels, embodiments of present invention provide forming a metal level **200** through a subtractive patterning process. As is demonstratively illustrated in FIGS.

1A, 1B, and 1C, a conductive material layer such as a metal layer **201** may be formed on top of a supporting structure **101** through, for example, a chemical vapor deposition (CVD) process, an atomic layer deposition (ALD) process, or a physical vapor deposition (PVD) process. The supporting structure **101** may be a part of the BSPDN structure, a part of the BEOL structure, a middle-of-line (MOL) structure, a FEOL structure, or a substrate such as a semiconductor substrate. For example, as one embodiment, the metal layer **201** may be formed at a backside of the supporting structure **101** which may be a substrate. The metal layer **201** may include conductive materials such as, for example, copper (Cu), aluminum (Al), cobalt (Co), ruthenium (Ru), tungsten (W), molybdenum (Mo), and other materials suitable for forming the metal level **200**.

(16) Embodiments of present invention provide patterning the metal layer **201** to form multiple metal lines such as, for example, a first metal line and/or a second metal line of the metal level **200**. The first and/or second metal lines may form at least part of an interconnect. More specifically, the first metal line may include a first metal strip **210** and may include, for example, a notch **212** at a side of the first metal strip **210**. The second metal line may include a second metal strip **220** and may include, for example, a pass-through **222** inside the second metal strip **220**. The first and second metal strips **210** and **220**, as being illustrated in FIGS. **2A, 2B, and 2C**, run horizontally to have a length along the horizontal direction. Embodiments of present invention further provide patterning the metal layer **201** to form a conductive wiring **211** that is at least partially inside the notch **212** but separated from the first metal strip **210**, and/or a conductive wiring **221** that is inside the pass-through **222** but separated from the second metal strip **220**.

(17) In one embodiment, the first metal strip **210** and/or the second metal strip **220** may have a width **W1** ranging from about 10 nm to about 100 nm, and the width **W1** is defined as a width measured vertically, as illustrated in FIG. **2A**, at a region without the notch **212** or the pass-through **222**. In the backside of the semiconductor substrate, since the first metal strip **210** and/or the second metal strip **220** are mainly used as metal lines for backside power distribution, such as for example as a part of the BSPDN structure, their widths are generally larger or bigger than metal lines of metal levels in the BEOL structure at the frontside of the semiconductor substrate.

(18) In one embodiment, the notch **212** formed at one side of the first metal strip **210** may have a depth **N1** into the first metal strip **210**, and the depth **N1** may be about 50 percent or less of the width **W1** of the first metal strip **210**. In another embodiment, the pass-through **222** in the second metal strip **220** may be significantly a rectangular shape, and the rectangular shape may have a width **P1**, along the width **W1** of the second metal strip **220**, that is about 50 percent or less of the width **W1** of the second metal strip **220**.

(19) In one embodiment, the conductive wirings **211** and **221** may be conductive vias that run in a direction vertical to the first and second metal strips **210** and **220** such as in a direction into/out of the paper, as is illustrated in FIG. **2A**, or in an up/down direction parallel to the paper, as are illustrated in FIGS. **2B and 2C**. In other words, conductive wirings **211** and **221** vertically pass through the first and second metal strips **210** and **220** and, as being described below in more details, are insulated from the first and second metal strips **210** and **220** respectively by dielectric materials.

(20) The first and second metal strips **210** and **220** and the conductive wirings **211** and **221** may be patterned through a lithographic patterning and etching process. More specifically, a hard mask may be formed to cover a portion of the metal layer **201**. The hard mask may have a mask pattern representing the first and second metal strips **210** and **220** and the conductive wirings **211** and **221**. Next, a selective etching process, such as a reactive-ion-etching (RIE) process, may be applied to remove or etch away portions of the metal layer **201** that are not covered by the hard mask to expose the underneath supporting structure **101**, thereby resulting remaining portions of the metal layer **201** to become the first and second metal strips **210** and **220** and the conductive wirings **211** and **221**.

(21) Furthermore, embodiments of present invention provide forming a dielectric layer **202**, such as

through a CVD, ALD, or PVD process, surrounding the first and second metal strips **210** and **220** and the conductive wirings **211** and **221**. As being demonstratively illustrated in FIGS. 3A, 3B, and 3C, the notch **212** becomes a notch **213** filled with material of the dielectric layer **202**, and the conductive wiring **211** is surrounded by the material of the dielectric layer **202** to be insulated from the first metal strip **210**. Similarly, the pass-through **222** becomes a pass-through **223** filled with material of the dielectric layer **202**, and the conductive wiring **221** is surrounded by the material of the dielectric layer **202** to be insulated from the second metal strip **220**.

(22) FIGS. 4A, 4B, and 4C to FIGS. 6A, 6B, and 6C are demonstrative illustrations of top views and cross-sectional views of a semiconductor structure in a process of manufacturing thereof according to one embodiment of a method of present invention. More particularly, embodiments of present invention provide forming a metal level **300** through a damascene patterning process. As is demonstratively illustrated in FIGS. 4A, 4B, and 4C, a dielectric material layer such as a dielectric layer **301** may be formed on top of the supporting structure **101** through a CVD, ALD, or PVD process. The dielectric layer **301** may be silicon-nitride (SiN), silicon-carbonitride (SiCN), silicon-carbide (SiC), aluminum-oxide (AlO.sub.2), aluminum-nitride (AlN), and other suitable materials.

(23) Embodiments of present invention provide forming one or more openings in the dielectric layer **301**, through a lithographic patterning and etching process. The one or more openings may include, for example, a first opening **310** for a first metal line and a second opening **320** for a second metal line that may run horizontally, as illustrated in FIGS. 5A, 5B, and 5C. In one embodiment, additional openings such as a third opening **311** and a fourth opening **321** may be formed in the dielectric layer **301** for a first conductive wiring **331** and a second conductive wiring **341** respectively. The formation of the first, second, third, and fourth openings **310**, **320**, **311**, and **321** transforms the dielectric layer **301** into a dielectric layer **302**, which surrounds the first, second, third, and fourth openings **310**, **320**, **311**, and **321**. The first, second, third, and fourth openings **310**, **320**, **311**, and **321** may be made by first forming a hard mask covering portions of the dielectric layer **301** and removing, through a selective etching process, portions of the dielectric layer **301** that are not covered by the hard mask to form the openings.

(24) The first opening **310** for the first metal line may include a notch **312** made of the material of the dielectric layer **302** that in-turn surrounds the third opening **311** for the first conductive wiring **331**. The second opening **320** for the second metal line may include a pass-through **322** made of the material of the dielectric layer **302** that in-turn surrounds the fourth opening **321** for the second conductive wiring **341**.

(25) Embodiments of present invention provide further filling the first, second, third, and fourth openings **310**, **320**, **311**, and **321** with a conductive material such as Cu, Co, Al, Ru, W, Mo, etc. to form the first metal line and the second metal line and the first and second conductive wirings. More particularly, as illustrated in FIGS. 6A, 6B, and 6C, the first metal line may include a first metal strip **330** and the first conductive wiring **331** may be at least partially inside the notch **312** and surrounded by the dielectric layer **302**. The second metal line may include a second metal strip **340** and the second conductive wiring **341** may be inside the pass-through **322** and surrounded by the dielectric layer **302**.

(26) The first metal strip **330** and the second metal strip **340** may have a width **W1**, the notch **312** may have a depth **N1**, and the pass-through **322** may have a width **P1** that are similar to those as being discussed above with reference to FIG. 2A. Moreover, the first and second conductive wirings **331** and **341** may be conductive vias, similar to conductive wirings **211** and **221**.

(27) FIGS. 7A, 7B, and 7C to FIGS. 10A, 10B, and 10C are demonstrative illustrations of top views and cross-sectional views of a semiconductor structure in a process of manufacturing thereof according to one embodiment of a method of present invention. More particularly, embodiments of present invention provide forming a metal level **400** through a combination of subtractive patterning process and a damascene patterning process. For example, a first and a second metal line may first be formed through a subtractive patterning process. A notch and/or a pass-through may be

formed, optionally together with the formation of the first and second metal lines. Next, a conductive wiring or via that is at least partially inside the notch and/or a conductive wiring or via that is fully inside the pass-through may be formed through a damascene patterning process. (28) More specifically, as illustrated in FIGS. 7A, 7B, and 7C, embodiments of present invention provide forming a conductive material layer such as a metal layer **401** on top of the supporting structure **101** through a CVD, ALD, or PVD process. The metal layer **401** may include conductive materials such as, for example, Cu, Al, Co, Ru, W, Mo, and other materials suitable for forming the metal level **400**.

(29) Embodiments of present invention provide patterning the metal layer **401** to form multiple metal lines such as, for example, a first metal line and a second metal line of the metal level **400**. For example, as being illustrated in FIGS. 8A, 8B, and 8C, the first metal line may include a first metal strip **410** and the second metal line may include a second metal strip **420**. In one embodiment, the first metal line may be formed to include a notch **411** at a side of the first metal strip **410**. In another embodiment, the second metal line may be formed to include a pass-through **421** inside the second metal strip **420**.

(30) However, embodiments of present invention are not limited in this aspect. For example, in one embodiment, the notch **411** and/or the pass-through **421** may be formed in a separate subtractive patterning process or a damascene patterning process after the first metal strip **410** and/or the second metal strip **420** is formed.

(31) As being demonstratively illustrated in FIGS. 9A, 9B, and 9C, a dielectric layer **402** may be formed, through a CVD, ALD, or PVD process, to surround the first metal strip **410** and the second metal strip **420**. The dielectric layer **402** may fill the opening of the notch **411** and the opening of the pass-through **421**, thus producing a notch **412** and a pass-through **422**, and to surround the first and second metal strips **410** and **420**. As being described above, in one embodiment, the notch **411** and/or the pass-through **421** may be created at this stage through a subtractive patterning process before the dielectric layer **402** is formed to surround the first and second metal strips **410** and **420**, and after the creation of the notch **411** and/or the pass-through **421** the dielectric layer **402** may be formed to fill the opening of the notch **411** and the opening of the pass-through **421** to produce a notch **412** and a pass-through **422**. The dielectric layer **402** may be SiN, SiCN, SiC, AlO.sub.2, AlN, and other suitable materials.

(32) Embodiments of present invention provide further forming a first and a second conductive wirings at least partially in the notch **412** and/or inside the pass-through **422** through a damascene patterning process. More specifically, a first opening **413** may be created in the dielectric layer **402** in the notch **412** and a second opening **423** may be created in the dielectric layer **402** inside the pass-through **422**. The first and second openings **413** and **423** may be created through a lithographic patterning and etching process.

(33) As being demonstratively illustrated in FIGS. 10A, 10B, and 10C, following the creation of the first and second openings **413** and **423**, embodiments of present invention provide filling the first and second openings **413** and **423** with a conductive material to form a first conductive wiring **414** and a second conductive wiring **424**. The first and second conductive wirings **414** and **424**, which may be conductive vias, are surrounded by the material of the dielectric layer **402** and thus insulated from the first metal strip **410** and the second metal strip **420** respectively. The first and second metal strips **410** and **420**, the notch **412**, and the pass-through **422** may have their respective widths and/or depth as those being described above with reference to FIG. 2A.

(34) FIGS. 11A, 11B, and 11C to FIGS. 14A, 14B, and 14C are demonstrative illustrations of top views and cross-sectional views of a semiconductor structure in a process of manufacturing thereof according to one embodiment of a method of present invention. More particularly, embodiments of present invention provide forming a metal level **500** through a combination of subtractive patterning process and a damascene patterning process. For example, a first and a second metal line may first be formed through a damascene patterning process. A notch and/or a pass-through may be

formed next, or optionally formed together with the formation of the first and second metal lines. Next, a conductive wiring or via that is at least partially inside the notch and/or a conductive wiring that is fully inside the pass-through may be formed through a damascene patterning process.

(35) More specifically, as illustrated in FIGS. **11A**, **11B**, and **11C**, embodiments of present invention provide forming a dielectric material layer such as a dielectric layer **501** on top of the supporting structure **101** through a CVD, ALD, or PVD process. The dielectric layer **501** may be SiN, SiCN, SiC, AlO.sub.2, AlN, and other suitable materials.

(36) Embodiments of present invention provide forming one or more openings in the dielectric layer **501**, through a lithographic patterning and etching process. The one or more openings may include, for example, a first opening **510** for a first metal line and a second opening **520** for a second metal line, as illustrated in FIGS. **12A**, **12B**, and **12C**, that may run horizontally. The formation of the first and second openings **510** and **520** transforms the dielectric layer **501** into a dielectric layer **502**, which surrounds the first and second openings **510** and **520**. The openings may be made by first forming a hard mask covering portions of the dielectric layer **501** and removing, through a selective etching process, portions of the dielectric layer **501** that are not covered by the hard mask to form the openings.

(37) The first opening **510** for the first metal line may include a notch **511** made of the material of the dielectric layer **502** and the second opening **520** for the second metal line may include a pass-through **521** made of the material of the dielectric layer **502**. However, embodiments of present invention are not limited in this aspect. For example, in one embodiment, the notch **511** and/or the pass-through **521** may be formed in a separate subtractive patterning process or a damascene patterning process.

(38) As being demonstratively illustrated in FIGS. **13A**, **13B**, and **13C**, embodiments of present invention provide further filling the first and second openings **510** and **520** with a conductive material such as Cu, Co, Al, Ru, etc. to form a first metal line and a second metal line. More particularly, the first metal line may include a first metal strip **530** and the second metal line may include a second metal strip **540**.

(39) Embodiments of present invention provide further forming a first and a second conductive wirings at least partially in the notch **511** and/or inside the pass-through **521** through a damascene patterning process. More specifically, a first opening **531** may be created in the dielectric layer **502** in the notch **511** and a second opening **541** may be created in the dielectric layer **502** inside the pass-through **521**. The first and second openings **531** and **541** may be created through a lithographic patterning and etching process.

(40) As being demonstratively illustrated in FIGS. **14A**, **14B**, and **14C**, following the creation of the first and second openings **531** and **541**, embodiments of present invention provide filling the first and second openings **531** and **541** with a conductive material to form a first conductive wiring **532** and a second conductive wiring **542**. The first and second conductive wirings **532** and **542** are surrounded by the material of the dielectric layer **502** and insulated from the first metal strip **530** and the second metal strip **540** respectively by the dielectric layer **502**. The first and second metal strips **530** and **540**, the notch **511**, and the pass-through **521** may have their respective widths and/or depth as those being described above with reference to FIG. **2A**.

(41) FIG. **15** is a demonstrative illustration of top view of a semiconductor structure according to embodiments of present invention. More particularly, embodiments of present invention provide a semiconductor structure that includes multiple metal levels such as, for example, a first metal level **600**, a second metal level **700**, and one or more metal levels. Moreover, each metal levels may include one or more metal lines. For example, the metal level **600** may include a first metal line that includes a first metal strip **610** with one or more notches such as notches **611** and **612** and one or more pass-throughs such as a pass-through **613**. The metal level **600** may also include a second metal line that includes a second metal strip **620** with one or more notches such as a notch **621** and one or more pass-throughs such as pass-throughs **622** and **623**. Further for example, the second

metal level **700** may include a first metal line that includes a first metal strip **710** with, for example, a notch **711**; a second metal line that includes a second metal strip **720** with, for example, a pass-through **721**; a third metal line that includes a third metal strip **730** with, for example, a notch **731**; and a fourth metal line that includes a fourth metal strip **740** with, for example, a pass-through **741**. The notches **611**, **612**, **621**, **711**, **731** and pass-throughs **613**, **622**, **623**, **721**, **741** may be filled with material of a dielectric layer **601**. The second metal strips **710**, **720**, **730**, and **740** of the second metal level **700** may be oriented to be orthogonal to the first metal strips **610** and **620** of the first metal level **600**. Conductive wirings such as conductive wirings **801**, **802**, **803**, **804**, **805**, and **806** may pass through the notches or pass-throughs while being surrounded by the dielectric materials of the dielectric layer **601** and insulated from the metal strips **610**, **620**, **710**, **720**, **730**, and **740**.

(42) In one embodiment, the notch **611** of the first metal strip **610** may overlap with the notch **711** of the first metal strip **710** and the conductive wiring **801** is at least partially in the notches **611** and **711**. In another embodiment, the notch **621** of the second metal strip **620** may overlap with the pass-through **721** of the second metal strip **720** and the conductive wiring **802** is inside the pass-through **721** and at least partially in the notch **621**. In yet another embodiment, the pass-through **613** of the first metal strip **610** may overlap with the notch **731** of the third metal strip **730** and the conductive wiring **803** is inside the pass-through **613** and at least partially in the notch **731**. In a further embodiment, the pass-through **623** of the second metal strip **620** may overlap with the pass-through **741** of the fourth metal strip **740** and the conductive wiring **804** is inside both the pass-through **623** and the pass-through **741**.

(43) In one embodiment, a notch such as the notch **612** of the first metal strip **610** and/or a pass-through such as the pass-through **622** of the second metal strip **620** may not overlap with metal lines or metal strips in another metal level such as the second metal level **700**.

(44) FIG. **16** is a demonstrative illustration of a flow-chart of a method of manufacturing a semiconductor structure according to embodiments of present invention. The method includes **(910)** providing a substrate with a frontside and a backside and at least one transistor structure at the frontside of the substrate, and forming a first metal level at a backside of the substrate by one of the following processes: **(920)** etching a layer of conductive material in a subtractive patterning process to form a metal strip with a notch at a side of the metal strip or a pass-through inside the metal strip, and **(930)** creating an opening in a dielectric material in the notch or in the pass-through and filling the opening with a conductive material to form a conductive wiring or via, or **(940)** etching a layer of dielectric material to create openings and filling the openings with a conductive material in a damascene patterning process to form a metal strip with a notch at a side of the metal strip or a pass-through inside the metal strip, and **(950)** creating an opening in a dielectric material in the notch or in the pass-through and filling the opening with a conductive material to form a conductive wiring or via, or **(960)** etching a layer of conductive material in a subtractive patterning process to form a metal strip with a notch at a side of the metal strip or a pass-through inside the metal strip, and a conductive wiring vertically passing through the metal strip at the notch or the pass-through, or **(970)** etching a layer of dielectric material to create openings and fill the openings with a conductive material in a damascene patterning process to form a metal strip with a notch at a side of the metal strip or a pass-through inside the metal strip, and a conductive wiring vertically passing through the metal strip at the notch or the pass through, and **(980)** forming a second metal level above the first metal level by forming another metal strip with a notch or a pass-through.

(45) It is to be understood that the exemplary methods discussed herein may be readily incorporated with other semiconductor processing flows, semiconductor devices, and integrated circuits with various analog and digital circuitry or mixed-signal circuitry. In particular, integrated circuit dies can be fabricated with various devices such as field-effect transistors, bipolar transistors, metal-oxide-semiconductor transistors, diodes, capacitors, inductors, etc. An integrated circuit in accordance with the present invention can be employed in applications, hardware, and/or electronic systems. Suitable hardware and systems for implementing the invention may include, but

are not limited to, personal computers, communication networks, electronic commerce systems, portable communications devices (e.g., cell phones), solid-state media storage devices, functional circuitry, etc. Systems and hardware incorporating such integrated circuits are considered part of the embodiments described herein. Given the teachings of the invention provided herein, one of ordinary skill in the art will be able to contemplate other implementations and applications of the techniques of the invention.

(46) Accordingly, at least portions of one or more of the semiconductor structures described herein may be implemented in integrated circuits. The resulting integrated circuit chips may be distributed by the fabricator in raw wafer form (that is, as a single wafer that has multiple unpackaged chips), as a bare die, or in a packaged form. In the latter case the chip may be mounted in a single chip package (such as a plastic carrier, with leads that are affixed to a motherboard or other high-level carrier) or in a multichip package (such as a ceramic carrier that has surface interconnections and/or buried interconnections). In any case the chip may then be integrated with other chips, discrete circuit elements, and/or other signal processing devices as part of either an intermediate product, such as a motherboard, or an end product. The end product may be any product that includes integrated circuit chips, ranging from toys and other low-end applications to advanced computer products having a display, a keyboard or other input device, and a central processor.

(47) The descriptions above have been presented for the purposes of illustration of various embodiments of present invention and they are not intended to be exhaustive and present invention are not limited to the embodiments disclosed. The terminology used herein was chosen to best explain the principles of the embodiments, practical application or technical improvement over technologies found in the marketplace, and to enable others of ordinary skill in the art to understand the embodiments disclosed herein. Many modifications, substitutions, changes, and equivalents will now occur to those of ordinary skill in the art. Such changes, modification, and/or alternative embodiments may be made without departing from the spirit of present invention and are hereby all contemplated and considered within the scope of present invention. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the spirit of the invention.

Claims

1. A semiconductor structure comprising a metal level, the metal level comprising: a metal strip including a notch at a side of the metal strip; and a conductive wiring, the conductive wiring being at least partially inside the notch; vertically passing through the metal strip; and being insulated from the metal strip by a dielectric material.
2. The semiconductor structure of claim 1, further comprising a substrate having a frontside and a backside and one or more transistor structures on the frontside of the substrate, wherein the metal level is at the backside of the substrate.
3. The semiconductor structure of claim 2, wherein the metal level is a first metal level; the metal strip is a first metal strip; and the notch is a first notch, further comprising a second metal level above the first metal level, the second metal level comprising a second metal strip including a second notch at a side of the second metal strip, wherein the conductive wire is at least partially inside the second notch and insulated from the second metal strip.
4. The semiconductor structure of claim 2, wherein the metal level is a first metal level; and the metal strip is a first metal strip, further comprising a second metal level above the first metal level, the second metal level comprising a second metal strip including a pass-through inside the second metal strip, wherein the conductive wire is inside the pass-through and insulated from the second metal strip.
5. The semiconductor structure of claim 4, wherein the second metal strip of the second metal level is oriented to be orthogonal to the first metal strip of the first metal level, and the conductive wiring

is a conductive via.

6. The semiconductor structure of claim 1, wherein the notch formed at the side of the metal strip has a depth into the metal strip, wherein the depth is 50 percent or less of a width of the metal strip.

7. The semiconductor structure of claim 1, wherein the metal strip has a width ranging from about 10 nm to about 100 nm.

8. A semiconductor structure comprising a metal level, the metal level comprising a metal strip including a pass-through inside the metal strip, wherein the pass-through is at least partially filled with a dielectric material.

9. The semiconductor structure of claim 8, wherein the metal level further includes a conductive wiring, the conductive wiring vertically passing through the pass-through and being insulated from the metal strip by the dielectric material.

10. The semiconductor structure of claim 9, wherein the metal level is a first metal level; the metal strip is a first metal strip; and the pass-through is a first pass-through, further comprising a second metal level above the first metal level, the second metal level comprising a second metal strip including a second pass-through inside the second metal strip, wherein the conductive wiring is inside the second pass-through and insulated from the second metal strip.

11. The semiconductor structure of claim 9, wherein the metal level is a first metal level; and the metal strip is a first metal strip, further comprising a second metal level above the first metal level, the second metal level comprising a second metal strip including a notch at a side of the second metal strip, wherein the conductive wiring is at least partially inside the notch and insulated from the second metal strip.

12. The semiconductor structure of claim 11, wherein the second metal strip of the second metal level is oriented to be orthogonal to the first metal strip of the first metal level, and the conductive wiring is a conductive via.

13. The semiconductor structure of claim 8, wherein the metal strip has a width ranging from about 10 nm to about 100 nm, and the pass-through inside the metal strip has a rectangular shape with a width of the rectangular shape no larger than 50 percent of the width of the metal strip.

14. A method comprising: providing a substrate with a frontside and a backside and at least one transistor structure on the frontside of the substrate; forming a metal level at the backside of the substrate, forming the metal level comprising forming a metal strip including a notch at a side of the metal strip or including a pass-through inside the metal strip, wherein the notch or the pass-through is at least partially filled with a dielectric material.

15. The method of claim 14, wherein forming the metal level further comprises forming a conductive wiring that vertically passes through the metal strip, wherein the conductive wiring is at least partially inside the notch or inside the pass-through and being insulated from the metal strip by the dielectric material.

16. The method of claim 15, wherein forming the metal strip comprises depositing a layer of conductive material on top of a substrate; forming a hard-mask pattern on top of the layer of conductive material; etching the layer of conductive material exposed by the hard-mask pattern to form the conductive wiring and the metal strip with the notch or the pass-through; and filling the notch or the pass-through with the dielectric material, the dielectric material thereby surrounding the conductive wiring.

17. The method of claim 15, wherein forming the metal strip comprises depositing a layer of dielectric material on top of a substrate; forming a hard-mask pattern on top of the layer of dielectric material; etching the layer of dielectric material exposed by the hard-mask pattern to create one or more openings in the layer of dielectric material; and filling the one or more openings with a conductive material to form the conductive wiring and form the metal strip with the notch or the pass-through.

18. The method of claim 14, further comprising creating an opening in the dielectric material in the notch or in the pass-through; and filling the opening with a conductive material to form a

conductive wiring, the conductive wiring vertically passing through the metal strip and being insulated from the metal strip by the dielectric material.

19. The method of claim 18, wherein forming the metal strip comprises depositing a layer of conductive material on top of a substrate; forming a hard-mask pattern on top of the layer of conductive material; etching the layer of conductive material exposed by the hard-mask pattern to form the metal strip with the notch or with the pass-through; and filling the notch or the pass-through with the dielectric material.

20. The method of claim 18, wherein forming the metal strip comprises depositing a layer of dielectric material on top of a substrate; forming a hard-mask pattern on top of the layer of dielectric material; etching the layer of dielectric material exposed by the hard-mask pattern to create one or more openings in the layer of dielectric material; and filling the one or more openings with the conductive material to form the metal strip with the notch or with the pass-through.
